

QUIJOTE scientific results – XII. Polarized synchrotron emission at the power spectrum level in the QUIJOTE-MFI wide survey

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ABSTRACT

We present results on the properties of the polarization of the diffuse synchrotron emission of the galaxy as a foreground of the cosmic microwave background (CMB) polarization. The present analysis makes use of the QUIJOTE 11 and 13 GHz data together with the WMAP and Planck data at higher frequencies. We find that the EE and BB power spectra of the synchrotron are well described by a power law with an index around -3.1 ± 0.1 and a ratio BB/EE around 0.20 ± 0.05 . By comparing different foreground models we find that there is no evidence for spatial variation of the spectral energy distribution (SED) of the polarized synchrotron at the power spectrum level. The SED is well described by a power law with indices $\beta_s^{EE} = -3.00 \pm 0.05$ and $\beta_s^{BB} = -3.01 \pm 0.15$ for the E - and B -modes, respectively. We find that the spatial correlation between the synchrotron and the dust is ~ 0.2 . Finally, in the light of these results, we investigate the level of contamination to the CMB B -modes due the synchrotron emission.

Key words: cosmology: observations – cosmic microwave background

1 INTRODUCTION

Cosmic microwave background (CMB) intensity anisotropies have been central to the development of modern cosmology, providing key measurements that have contributed to establishing the standard cosmological model with remarkable accuracy (Smoot et al. 1992; Bennett et al. 2013; Planck Collaboration et al. 2020a). As observational sensitivity has steadily improved, the emphasis has increasingly shifted towards the study of CMB polarization anisotropies, as they enable searches for primordial B modes, whose detection would provide direct evidence for a stochastic background of gravitational waves generated during cosmic inflation (Kamionkowski et al. 1997; Zaldarriaga & Seljak 1997; Kamionkowski & Kovetz

2016). Over the past decade, significant progress has been made in constraining the amplitude of such signals, with the most stringent upper limits on the tensor-to-scalar ratio currently provided by the BICEP/Keck experiment (Ade et al. 2021), further strengthened by the inclusion of Planck data (Tristram et al. 2022), to the level of $r < 0.032$ at 95 per cent confidence level.

Looking ahead, a new generation of ground-based observatories, including the Simons Observatory (Ade et al. 2019) and CMB-S4 (Abazajian et al. 2022) in the southern hemisphere, and QUIJOTE (Rubio-Martín et al. 2012b), LSPE (Addamo et al. 2021) and GroundBIRD (Honda et al. 2020) in the northern one, together with dedicated space missions such as LiteBIRD (LiteBIRD Collab-

oration et al. 2023) and PICO (Hanany et al. 2019) aim to improve sensitivity to primordial B-modes by more than an order of magnitude, placing unprecedented demands on instrumental performance, systematic control and foreground subtraction. This paper focuses on this third aspect.

The main diffuse polarized contaminants are the synchrotron emission (which dominates at frequencies below ~ 70 GHz) and the thermal dust emission (which is the dominant mechanism at high frequencies) (Akrami et al. 2018a). Anomalous microwave emission (AME) has been detected in intensity, but no polarization has been measured up to date (Rubiño-Martín et al. 2012a; Dickinson et al. 2018), with current upper limits below the level of one per cent (González-González et al. 2025). According to the concordance model, the peak of the CMB B -mode power spectrum – or CMB BB power spectrum – is expected to be at intermediate angular scale around multipole $\ell \approx 80$ (e.g. Lewis et al. 2000). In this range of scales, there is no frequency domain or sky region where the sum of the synchrotron and thermal dust foregrounds is subdominant with respect to a CMB B -mode signal if it has an amplitude of $r = 10^{-3}$ (Planck Collaboration et al. 2016; Krachmalnicoff et al. 2018).

Any robust determination of the tensor-to-scalar ratio, r , will require an accurate separation of the cosmological signal from these Galactic foreground emissions (Remazeilles et al. 2016; Errard & Stompor 2018). Recent studies have revealed increasing complexity in both the spectral and spatial properties of Galactic dust and synchrotron emission, posing significant challenges for component separation (Choi & Page 2015; Planck Collaboration et al. 2017; Krachmalnicoff et al. 2018; Fuskeland et al. 2021; Weiland et al. 2022; de la Hoz et al. 2023). This complexity is particularly relevant in the case of polarised synchrotron emission. Owing to limited sensitivity in the low-frequency channels of *Planck* and *WMAP*, polarised synchrotron emission cannot be robustly detected at intermediate and high Galactic latitudes. As a result, the constraints on the spectral behavior of polarized synchrotron emission in these regions remain limited when using satellite data. In this context, dedicated observations at lower frequencies are required to complement existing satellite data and to improve foreground modeling to the level necessary for future B-mode polarization studies.

Recently, three radio surveys are providing key and complementary information in polarization: the S-PASS experiment, which observed the sky from the South hemisphere at 2.3 GHz (Carretti et al. 2019); the C-Band All-Sky Survey (C-BASS) experiment, which will cover the full sky at 5 GHz (Jones et al. 2018), and the QUIJOTE (Q-U-I JOint TEnerife) experiment, a multi-frequency microwave survey carried out at the Teide Observatory, covering the 10–40 GHz frequency range with angular resolutions between 18 arcmin and one degree (Rubiño-Martín et al. 2012b).

Previous studies seem to indicate that the polarized synchrotron β_s has a mean value close to -3.15 (Fuskeland et al. 2014; Akrami et al. 2018b). With the inclusion of S-PASS data, this average value was confirmed, and variations at the level of $\beta_s = -3.22 \pm 0.08$ were found between 2.3 and 33 GHz (Krachmalnicoff et al. 2018). QUIJOTE results published so far on the polarized synchrotron properties are mostly based on real-space component separation methods. Using a parametric approach on a pixel-by-pixel basis, de la Hoz et al. (2023) found an average value of the synchrotron spectral index of $\beta_s = -3.08 \pm 0.13$ at two-degree angular scales. This result is consistent with the average synchrotron spectral index measured between the MFI 11 GHz and *WMAP* 23 GHz channels of $\beta_s = -3.09 \pm 0.14$ (Rubiño-Martín et al. 2023). More recently, Adak et al. (2025) applied a semi-blind component separation method and obtained an average value of $\beta_s = -3.11 \pm 0.21$. All these

results hint towards spatial variations of the polarized synchrotron spectral index larger than the values reported in current models by a factor of 2–3 (e.g. Krachmalnicoff et al. 2018; de la Hoz et al. 2023). Quantifying the spatial variability of β_s is crucial to estimate the proper amount of synchrotron power at the CMB frequencies observed by current and forthcoming experiments.

This paper is part of a series associated with the QUIJOTE MFI wide survey (Rubiño-Martín et al. 2023), which covers approximately $29,000 \text{ deg}^2$ including most of the northern sky, in four frequency bands centered at 11.1, 12.9, 16.8, and 18.8 GHz. The survey achieves an angular resolution of approximately one degree and sensitivities in linear polarization in the range $35\text{--}40 \mu\text{K}$ per degree beam. This range of QUIJOTE-MFI frequencies is sufficiently close to the *Planck* and *WMAP* bands to mitigate depolarization effects due to Faraday rotation, while at the same time extending the frequency coverage and providing a longer lever arm to constrain the synchrotron spectral energy distribution (SED).

The analysis carried out in this paper focuses on the characterization of the spectral properties of the polarized synchrotron emission using harmonic space tools, following a similar methodology to the one used in Choi & Page (2015); Planck Collaboration et al. (2017); Krachmalnicoff et al. (2018); Martire et al. (2022). By combining QUIJOTE-MFI, *Planck* and *WMAP* data, we investigate if the spatial variation of β_s is significant enough to induce a departure of the synchrotron power spectrum SED (E - and B -modes) from a pure power law. The dust is the dominant polarized foreground above 70 GHz and is still a significative signal below (Adam et al. 2016). We also include the frequencies of the polarized channels of the *Planck* high frequency instrument (HFI) in order to better remove the dust from the data during the analysis, and for a better characterization of the synchrotron-dust correlation. In our analysis, we assume that both free-free and AME have negligible polarization fractions.

In the general case, the variation over the sky of the SED of an emission can not be entirely taken into account in a power spectrum analysis. The usual way to deal with spatially varying SEDs is to assign more than one component with fixed SED to the physical emission. The SED of each component can be obtained with non-parametric methods such as SMICA (Cardoso et al. 2008) or ILC (Bennett et al. 1992; Delabrouille et al. 2011), or with parametric methods (e.g. Eriksen et al. 2008; de la Hoz et al. 2023). A parametrised version of the SED can be obtained by expanding the SED with respect to its parameters that vary across the sky. In the assumption that the variation of the SED parameters over the sky is small, a small number of components is needed for a good enough description, i.e. within the noise levels achieved by current experiments (Chluba et al. 2017). In the paper, we will study the number of components necessary to describe both the synchrotron and the dust contained in the data.

The paper is organised as follows. In Sec. 2 we give a description of the data used for the analysis, focusing on the power spectra. In Sec. 3 we describe the methodology, including the foreground models and the fitting procedure. In Sec. 4 we present the results on the number of components needed to describe the foregrounds. In Sec. 5 we present the results on the properties of the polarized synchrotron. We finally discuss and conclude in Sec. 6.

2 MAPS AND POWER SPECTRA

In our analysis, we consider polarization data from three experiments: QUIJOTE-MFI, *WMAP*, and *Planck*. We exploit the broad

frequency coverage to provide a strong lever arm for constraining the synchrotron spectral energy distribution (SED), while the higher-frequency channels help to disentangle synchrotron and dust emission and to characterise their cross-correlation. All maps use a HEALPix¹ (Górski et al. 2005) pixelization. Details of each survey, together with the data combinations adopted in our analysis, and the codes used for power spectra computations are presented in the following subsections.

2.1 Maps

2.1.1 WMAP and Planck data

For consistency with the analyses of Krachmalnicoff et al. (2018) and Martire et al. (2022), we use only the 30 GHz channel of the *Planck* Low Frequency Instrument (LFI). However, unlike in those previous studies, we include also the four *Planck* High Frequency Instrument (HFI) channels at 100, 143, 217, and 353 GHz to improve the separation between synchrotron and dust emission and to investigate their cross-correlation properties. Our results are based on the *Planck* PR4 data release (NPIPE, Planck Collaboration et al. 2020c). All the maps were downloaded from the Planck legacy archive². Ninety-nine (numbers 201st to the 299th) of the end-to-end (E2E) noise simulations described in Planck Collaboration et al. (2020c) are used to estimate noise power spectra of the maps.

As in Krachmalnicoff et al. (2018), we use the *WMAP* nine-year polarization maps of the two lowest bands, i.e. K and Ka with central frequencies 23 and 33 GHz (Bennett et al. 2013). We combine the differential assembly (D/A) maps for those bands to construct a 1–4 year map and a 5–9 year map, which are then used to produce a half-difference map, scaling the half-maps so that the half-difference have the same amount of noise than the full nominal maps. These half maps are used to perform null tests and estimate the noise power spectra. All maps were downloaded from the LAMBDA website³.

All maps are converted in units of antenna temperature using the so-called unit conversion factors, and they are also colour corrected using an average SED for the synchrotron and dust components and the *fastcc*⁴ code (Peel et al. 2022). The corresponding factors are presented in Table 1.

2.1.2 QUIJOTE data

We use the public⁵ QUIJOTE-MFI wide survey DR1 maps in polarization at 11.1 and 12.9 GHz (Rubiño-Martín et al. 2023). We do not use the 17 and 19 GHz maps since they have a low signal to noise ratio in the diffuse regions and do not reduce the uncertainty on the target parameters of the paper.

The public half-maps are used to estimate the noise power spectra at 11 and 13 GHz. The half-difference of the two half-maps can be regarded as a single realization of the instrumental noise. However, by computing its power spectrum and averaging over wide ranges of multipoles, the resulting estimate provides a realistic approximation to the true noise power spectrum, as shown in Sect. 7 of Rubiño-Martín et al. (2023). Both maps are converted in units of antenna temperature and colour corrected using the factors of Table 1.

¹ <http://healpix.sourceforge.net>

² <http://pla.esac.esa.int/pla/#home>

³ <https://lambda.gsfc.nasa.gov>

⁴ <https://github.com/mpeel/fastcc>

⁵ QUIJOTE-MFI DR1: doi.org/10.26698/quijote-mfi-dr1

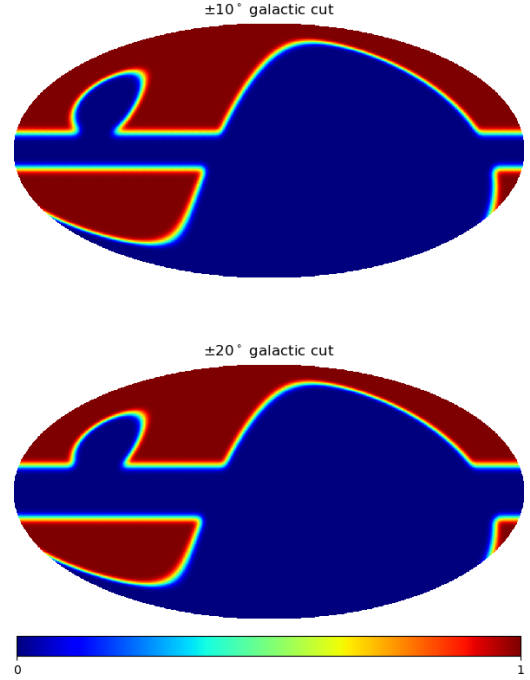


Figure 1. Example of two of the masks used in this analysis, corresponding to the combination of the QUIJOTE default analysis masks, and the galactic cuts $|b| > 10^\circ$ (top) and $|b| > 20^\circ$ (bottom). Here we consider three galactic cuts, namely 10, 15 and 20 degrees. The sky fractions after apodisation are 32, 28, 25 %, respectively. See text for further details.

2.1.3 Data set combinations

We consider three different combinations of the maps described in Sects. 2.1.1 and 2.1.2, defined as follows:

- (i) *WM+Pl*: includes the *WMAP* nine-year K and Ka band maps, together with the *Planck* full-mission maps at 30, 100, 143, 217, and 353 GHz, for a total of $N_f = 7$ frequency channels;
- (ii) *WM+Pl+QJ*: extends the *WM+Pl* set by including the QUIJOTE-MFI maps at 11 and 13 GHz, resulting in a total of $N_f = 9$ frequency channels;
- (iii) *NULL*: includes the same $N_f = 9$ frequency channels, but using half-difference maps for *WMAP* and QUIJOTE-MFI, and one noise realization for each *Planck* frequency channel.

The *WM+Pl* and *WM+Pl+QJ* combinations are used to quantify the impact of including QUIJOTE-MFI data in the analysis. The *NULL* set is employed to perform visual null tests, in particular to assess whether the adopted prior probabilities are overly informative.

2.2 Angular power spectra

Our analysis focuses on low and intermediate Galactic latitudes, using three masks with different sky fractions. These masks are constructed by combining the default QUIJOTE analysis masks (Rubiño-Martín et al. 2023) with Galactic latitude cuts of the form $|b| > b_{\text{cut}}$, where $b_{\text{cut}} = 10^\circ, 15^\circ$, and 20° . In this way, we explicitly exclude a band around the Galactic plane, as this work is focused on diffuse emission regions. For the power spectrum estimation, all masks are apodized using a Gaussian smoothing kernel with a full width at half maximum of 5° . After apodization, the resulting sky fractions are $f_{\text{sky}} = 32\%, 28\%,$ and 25% , respectively. The regions

Table 1. Colour correction (CC) and unit conversion (UC) factors adopted in this paper. CC factors are computed using `fastcc` for an average synchrotron spectrum with $\beta_s = -3.2$ and a modified black-body for the dust with $\beta_d = 1.55$ and $T_d = 19.6$ K. UC factors convert K_{CMB} to K_{RJ} , and are computed consistently with the definition of CC. For QUIJOTE-MFI and LFI, they correspond to the monochromatic conversion at the central frequency. For HFI, they follow the corresponding convention including the band-pass shift correction.

Frequency (GHz)	MFI 11.1	MFI 12.9	WMAP 22.8	WMAP 33	LFI 28.4	HFI 100	HFI 143	HFI 217	HFI 353
CC	0.964	0.997	0.946	0.969	0.999	0.993	0.993	0.924	0.907
UC	0.997	0.996	0.987	0.972	0.979	0.788	0.590	0.333	0.075

retained for the analysis are shown in red in Fig. 1 for the two cases with the lowest and the largest sky coverage.

We compute and use the auto-spectra of all frequencies and the cross-power spectra between all frequencies of the sets described in 2.1.3. The spectra are computed using the PolSpice algorithm (Chon et al. 2004), which corrects the spectra for most of the mask effects. As an internal consistency check, we compared the spectra we obtained with PolSpice and those obtained with NaMaster (Alonso et al. 2019), and we did not find any significant differences after binning the spectra. Loss of power due to the pixelisation is compensated by dividing by the HEALPix window function. The WMAP and Planck spectra are beam corrected using their respective instrument models. The QUIJOTE-MFI spectra are beam corrected using the beam transfer functions provided in their public data release

All spectra are binned in $N_{\text{bin}} = 5$ groups of $\Delta\ell = 20$ multipoles each, from $\ell = 30$ to $\ell = 129$, using the following weighted sum:

$$C_b = \sum_{\ell} w_{b\ell} C_{\ell}, \quad (1)$$

where b denotes the bin number, C_{ℓ} represents any auto- or cross-spectrum and $w_{b\ell} \propto (2\ell + 1)$ for $\ell_{\min}(b) \leq \ell \leq \ell_{\max}(b)$ and $w_{b\ell} = 0$ elsewhere and, for each b , $\sum_{\ell} w_{b\ell} = 1$. This weighting scheme is motivated by the form of the binned likelihood used for the analysis (see Sec. 3.4 below). The centre multipole of each bin is noted ℓ_b . In the rest of the paper, a capital letter with ℓ as an index (as on the right-hand side of Eq. 1) denotes an unbinned power spectrum and a capital letter with b as an index (as on the left-hand side of Eq. 1) denotes a binned power spectrum. Throughout the paper, angular power spectra in figures are depicted with the usual band power $D_{\ell} = \ell(\ell + 1)C_{\ell}/(2\pi)$.

Fig. 2 shows part of the auto- and cross-spectra used for the analysis. Each plot shows the spectra of the *WM+PI+QJ* and *NULL* sets. The spectra of the *NULL* set can be considered as the spectra of one realization of the instrumental noise. The absolute value of the *NULL* cross-spectra between QUIJOTE-MFI and WMAP or Planck represents an estimation of the standard deviation of the noise chance correlation. For QUIJOTE-MFI data, due to the fact that the frequency bands of the same horn share the same amplifier, the noise is correlated between 11 and 13 GHz. This is accounted in the null test maps adopted in the analysis. Finally, we note that QUIJOTE-MFI, when cross correlated with Planck and WMAP, provides high signal-to-noise information about the polarized synchrotron. In particular, the synchrotron-dust correlation is detected up to multipole $\ell \sim 100$.

The *TT*, *TE*, *TB* and *EB* cross-spectra are not considered in the present paper, and *E*- and *B*-modes are treated separately.

3 METHOD

This section describes the methodology adopted in this paper to estimate the foreground parameters. All analyses are performed in harmonic space, following an approach similar to that used in Krachmalnicoff et al. (2018) and Martire et al. (2022). The SEDs of the polarized foregrounds and the CMB are modeled using a parametric framework, and the E- and B-mode auto- and cross-power spectra are fitted to recover the key parameters describing the spectral behaviour of synchrotron and dust emission, their angular dependence, and their mutual cross-correlation.

3.1 Foreground model

We assume that the foregrounds consist of only polarized synchrotron and thermal dust emissions. As in previous works (e.g. Akrami et al. 2018a; Krachmalnicoff et al. 2018; Martire et al. 2022), we assume that the level of linear polarization of free-free emission and AME is negligible (e.g. Dickinson et al. 2018; González-González et al. 2025).

3.1.1 SED expansion and models

For each of the synchrotron and the dust, we consider two models. The first model is a single component model, i.e. the SED of the foreground does not vary spatially. This corresponds to a pure power law with spatially constant spectral index for the synchrotron, and a single modified blackbody with constant dust temperature and spectral index for the thermal dust emission. The second model is a two-component model where each component has a non spatially varying SED.

In the two-component model, the first component has the usual SED of the foreground, while the second component is a first order correction due to possible small variations of the spectral indices of the foregrounds. The temperature of the dust is always supposed to be fixed to a value of $T_{\text{dust}} = 19.6$ K [TBC] **FV:** I am not sure what needs to be confirmed, if it is the value of T_d used in the analysis then i can confirm this is the value that is used (link to the code <https://github.com/flavien-vansyngel/component-separation-on-quijote/blob/eaddceb76395f7479a7a0e22c72da91c1ce019b8/2-analysis/params.R#L30>), if it is the fact that only β_d (and not T_d) is allowed to vary on the sky then it can be seen in Eq. 8.

These two components correspond to the first two terms in the SED moment expansion presented in Chluba et al. (2017). This expansion allows to separate the spatial and spectral dependence of the foregrounds and treat one complex emission as several simpler components. Indeed, we can write the frequency dependence $\nu^{\beta(\mathbf{n})}$, which is different for each line of sight $\mathbf{n}$, as

$$\nu^{\beta(\mathbf{n})} = \nu^{\beta_0 + \delta\beta(\mathbf{n})} \simeq \nu^{\beta_0} + \delta\beta(\mathbf{n})\nu^{\beta_0} \log(\nu) \quad (2)$$

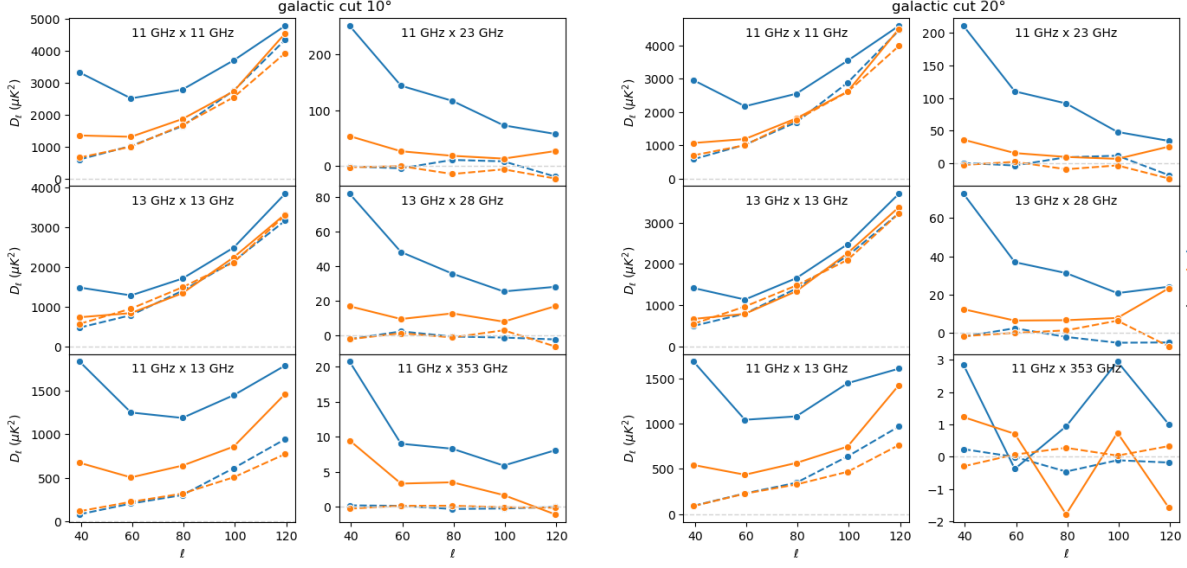


Figure 2. Part of the auto- and cross-spectra used in the analysis Left/right: 10/20 degrees galactic cut. In each panel, the left plots show the auto- and cross-spectra of 11 and 13 GHz of Horn 3 of QUIJOTE-MFI and the right plots show the cross-spectra between QUIJOTE 11 GHz and (from top to bottom) WMAP K band 23 GHz, LFI 30 GHz and HFI 353 GHz.

where β_0 is the mean of the $\beta(\mathbf{n})$ map and $\delta\beta(\mathbf{n})$ is a zero mean map. Thus, assuming $\delta\beta(\mathbf{n}) \ll 1$, we can approximate a line-of-sight dependent SED $a(\mathbf{n}, \nu)$ of a map $s(\mathbf{n})$ as the sum of two maps whose SED are fixed across the sky as follows

$$s(\mathbf{n}) a(\mathbf{n}, \nu) \simeq s_1(\mathbf{n}) a_1(\nu) + s_2(\mathbf{n}) a_2(\nu). \quad (3)$$

This decomposition can be applied to either total intensity maps, E - or B -mode maps.

3.1.2 Sets of composite models

As described in the previous subsection, in this paper we will explore whether the data prefers the single component description for synchrotron and/or dust, or the two-component model for any of the two cases. The total number of synchrotron and dust components and the CMB (for E -modes) is noted as N_c . We can thus build four different models, depending on whether the foregrounds are each described by one or two components. A model selection will sort out the number of degrees of freedom needed to best describe the foregrounds. In what follows, the notation $spdq$ refers to a model with a p -component synchrotron and a q -component dust (e.g. $s2d1$ stands for a 2 component synchrotron and 1 component dust and $N_c = 3$ without the CMB). As stated above, we only consider models with $p = 1, 2$ and $q = 1, 2$.

In the four possible $spdq$ models, the values of the power spectra of the components are a priori independent from multipole bin to multipole bin, and thus are fitted separately in each bin. However, in addition to the four models, a model noted $sfd1$ is used to fit a multipole spectral index for synchrotron, so we can compare with the literature values (Krachmalnicoff et al. 2018; Martire et al. 2022). For this $sfd1$ model, the dust is still a one-component dust with power spectra values independent from one bin to another, while the synchrotron is a one-component synchrotron but with a power spectrum parametrised as

$$S_\ell^{\text{synch.} \times \text{synch.}} = A_s \left(\frac{\ell}{80} \right)^\alpha, \quad (4)$$

using the same pivot scale of $\ell = 80$ as in Krachmalnicoff et al. (2018). Then instead of fitting for the value of the power spectrum of each bin (like in the $s1d1$ model for example), only the overall amplitude A_s and the index α are fitted in this case. As shown in 3.3, the amplitude parameter A_s is normalized at the pivot frequency of 23 GHz.

3.2 CMB model

In this work, the CMB E -modes are fitted along with the foregrounds. Thus we have additional parameters corresponding to the amplitudes of the EE power spectrum in each multipole bin. The SED of the CMB E -modes is kept fixed to the SED of the CMB anisotropies. We assume that the CMB does not contain any B -modes.

3.3 Combined data model

The data model used in this paper is a sum of two foregrounds (dust and synchrotron), CMB and noise. Thus for each multipole mode (ℓ, m) and for each polarization mode E or B (noted as X), the data vector can be decomposed as follows:

$$d_{\ell m}^X = A s_{\ell m}^X + n_{\ell m}^X. \quad (5)$$

The piece of data $d_{\ell m}^X$ is a vector whose entries represent the N_f frequencies. The component vector $s_{\ell m}^X$ has size N_c and each entry represents either the CMB (for EE only) or one component of the foregrounds, i.e. the spherical harmonics transform of the s_1 and possibly s_2 maps of Eq. 3). The number of component N_c depends on the foreground model chosen among those presented in Sec. 3.1.2. The $N_f \times N_c$ matrix A is the mixing matrix and gathers the SEDs of the galactic emissions and of the CMB, in units of K_{RJ} . Explicitly, the columns of the mixing matrix A contains the following elements for the CMB:

$$a(\nu) = x_\nu^2 \exp(x_\nu) / (\exp(x_\nu) - 1)^2, \quad (6)$$

these for the synchrotron:

$$\begin{aligned} a_1(\nu) &= (\nu/23)^{\beta_s} \\ a_2(\nu) &= (\nu/23)^{\beta_s} \log(\nu/23) \end{aligned} \quad (7)$$

and finally these for the thermal dust:

$$\begin{aligned} a_1(\nu) &= (\nu/353)^{\beta_d+1} (\exp(y_{353}) - 1) / (\exp(y_\nu) - 1) \\ a_2(\nu) &= (\nu/353)^{\beta_d+1} \log(\nu/353) (\exp(y_{353}) - 1) / (\exp(y_\nu) - 1), \end{aligned} \quad (8)$$

where $x_\nu = h \nu 10^9 / (k T_{\text{CMB}})$, $y_\nu = h \nu 10^9 / (k T_d)$, ν is given in GHz, and h and k are the Planck and Boltzmann constants, respectively. The spectral dependence of the synchrotron is encoded in its spectral index β_s , while the spectral index for the thermal dust SED is given by β_d . For the synchrotron amplitude, we are using the pivot frequency of 23 GHz, while for the dust amplitude, we use 353 GHz. **JARM: We are missing here the definition of the correlation dust-synch, ρ .** **FV: added below**

The vector $n_{\ell m}^X$ represents the instrumental noise in each channel and has same size as $d_{\ell m}^X$. The decomposition detailed in Sec. 3.1.1 allows to separate between frequency dependence (the mixing matrix) and the spatial dependence (the component vector). The power spectrum of the data model of Eq. 5, for either E - or B -modes, is then given by, after binning,

$$C_b(\theta) = A S_b A^T + N_b, \quad (9)$$

where superscript T denotes the transposition, the covariance matrix S_b gathers the binned power and cross spectra of the components and the covariance matrix N_b is the noise covariance.

FV: definition of ρ The correlation between any two components i and j among any of the synchrotron or dust components is $\rho_b^{i \times j} = S_b^{i \times j} / \sqrt{S_b^{i \times i} S_b^{j \times j}}$. In the result section, only the correlation between the synchrotron and the dust of the s1d1 model is of interest and is simply noted ρ_b .

The diagonal elements of N_b are computed using the maps described in Sec. 2.1.1 and 2.1.2 and then binned. All off-diagonal elements of N_b are assumed to be 0 except the covariance between the 11 and 13 GHz channels of QUIJOTE-MFI, which is fixed to the cross power spectrum between the half-difference maps, as explained above.

The vector θ in Eq. 9 is the set of the free parameters of the model, i.e. all entries of the binned signal covariance matrix S_b (except the correlations between the CMB and the foregrounds that are kept fixed to 0), the mean power law indices β_s and β_d of the SED of the synchrotron and the dust emissions. The special case of the foreground model sfd1 is detailed in the next subsection.

3.4 Fitting procedure

The method consists in building a posterior distribution over the set of model parameters θ and then sampling from the posterior possible values of θ . The posterior is the product between a parametrised version of the SMICA likelihood (Cardoso et al. 2008) and some prior distribution over θ . This method is a partial sky version of the blind Bayesian component separation presented in Vansyngel et al. (2016). The inference is performed by sampling the posterior using the `stan` language (Stan Development Team 2024) through the `rstan` interface (Stan Development Team 2025).

The likelihood, for either E - or B -modes, is a Wishart distribution – a generalisation to matrices of the chi-squared and gamma

Table 2. Bayes factors in favour of the sfd1 model against the other four models when using the dataset e WM+PI+QJ on galactic cut $\pm 10^\circ$, for E - and B -modes, expressed in base-10 logarithm. The trend is very much similar on the two other galactic cuts and/or with the dataset WM+PI.

	s1d1	s2d1	s1d2	s2d2
EE	5.13	11.9	16.5	23
BB	4.58	10.5	11	15.9

distributions – and can be written as follows

$$\mathcal{L}(d | \theta) \propto \prod_b |C_b(\theta)|^{-\frac{k_b}{2}} \exp \left[-\frac{k_b}{2} \text{Tr} \left\{ C_b(\theta)^{-1} \hat{C}_b(d) \right\} \right] \quad (10)$$

where the index b runs over the number of multipole bins, Tr is the trace operation, $\hat{C}_b(d)$ and $C_b(\theta) = A S_b A^T + N_b$ are the binned power spectra of the data and of the data model respectively. The number k_b is the number of degrees of freedom contained in bin b . We have $k_b = (2\ell_b + 1) \Delta \ell f_{\text{sky}}$ where ℓ_b is the central multipole of bin b .

The prior distributions on the dust and synchrotron emissivity indices are Gaussian distributions centered around the usual measured values: $\beta_s \sim \mathcal{N}(-3, 1)$ and $\beta_d \sim \mathcal{N}(1.6, 1)$. The covariance of the foregrounds S_b^{fgd} is reparametrised so that the sampled parameters are the square roots of the auto-spectra of the foregrounds $\sqrt{F_b}$ and their Pearson correlation matrix $R_b = L_b L_b^T$, i.e. $S_b^{\text{fgd}} = \text{diag}(\sqrt{F_b}) L_b L_b^T \text{diag}(\sqrt{F_b})$. This reparametrisation is meant to increase the efficiency of the sampling process. The prior distribution on $\sqrt{F_b}$ is a flat distribution and the prior on R_b is a uniform distribution, which in practice is implemented by using the Cholesky LKJ distribution of the `stan` language on L_b . The prior on the CMB E -modes is very informative in order to mimic a removal of the *Planck* best fit (Akrami et al. 2018c) from the data spectra and allowing some uncertainty in the process that will propagate to the uncertainties of all the other parameters. The inverse-gamma distribution is a natural choice to describe a variance. The parameters of the prior distribution on the CMB E -modes are tuned so that the mode is the *Planck* 2018 best fit power spectrum C_b^{Planck} and that the variance is very close to the cosmic variance in a partial sky, i.e. $S_b^{\text{CMB}} \sim \gamma^{-1} \left(k_b, (k_b + 1) C_b^{\text{Planck}} \right)$.

Regarding the sfd1 model, the set of parameters θ of Eq. 10 does not contain the N_b values of $\sqrt{F_b}$ for the synchrotron but instead contains the parameters A_s and α of Eq. 4. Both A_s and α have a flat prior distribution.

4 SELECTION OF THE FOREGROUND MODEL

In this section we first select the model that best fits the data among those presented in Sec. 2.1.3 and then we evaluate the goodness of fit of the best model.

4.1 Model relative selection

In order to select the most adapted foreground model among those presented in Sec. 3.1.2, a model selection is performed using Bayes factors (Kass et al. 1995).

For each model, we perform an inference of the parameters using the data of the dataset *WM+PI+QJ*. Then we compute the evidences of each inference and the Bayes factors between each pair of models with the R package `bridgesampling` (Gronau et al.

2020). The Bayes factor is a standard tool in Bayesian inference to compare two models. It can be interpreted as an odds ratio between the models. We consider the null hypothesis to be the simplest model we defined, i.e. *sfd1*, and through Bayes factor we test if we need to add any complexity to the data model presented in Sec. 3. Table 2 shows the ratios of the evidence of the *sfd1* model over the evidence of the other four models.

The Bayes factors are overwhelmingly in favour of the *sfd1* model. Only one component for the synchrotron and the dust are necessary to explain the data, meaning that the variability of the SED of the polarized synchrotron is not detected at the power spectrum level in the considered sky map and for the data combination *WM+PI+QJ*. The *sfd1* foreground model is therefore the model we will use in the remaining part of the paper to present the results of our analysis.

4.2 Goodness of fit

To quantitatively assess the goodness of fit, we first maximise the likelihood in Eq. 10 with respect to the parameters of the *sfd1* model and we note $\theta_m = \arg \max_{\theta} \mathcal{L}(d | \theta)$. The vector of parameters θ_m is then used to compute the log-likelihood ratio statistics

$$D = -2 \log \left[\frac{\mathcal{L}(d | \theta_m)}{\mathcal{L}(d | \tau_m)} \right] = \sum_b D_b \quad (11)$$

where τ_m is the set of parameters of the most general data model possible τ that maximises the likelihood. In this model τ , $C_b(\tau)$ is simply a $N_f \times N_f$ covariance without constraint and thus we have $C_b(\tau_m) = \hat{C}_b(d)$. According to Wilks' theorem (Wilks 1938), the quantities D_b tend to be distributed as a chi-squared distribution with a number of degrees of freedom n , which is the number of degrees of freedom in a N_f covariance, i.e. $N_f(N_f+1)/2 = 45$, minus the number of parameters per multipole of the model, i.e. the dust power and the dust-synchrotron correlation for each multipole, and the amplitude and index of the synchrotron power spectrum and the SEDs indices β_s and β_d per multipole bin, i.e. $2+(2+2)/N_{\text{bin}} = 2.8$. The CMB is not counted as a free parameter, since it is very much constrained by the priors. Thus the number of degrees of freedom of the chi-squared distribution is $n = 42.2$. We can then compute a goodness of fit metric from the log-likelihood ratios interpreted as chi-squared values. We choose the principle of total evidence (PTE), which evaluates the probability of the occurrence of a larger deviation from the model of Eq. 9 than what was observed in the data (e.g. Arrasmith et al. (2019)). In practice, PTE consists in evaluating the survival function of the chi-squared distribution of 42.2 degrees of freedom at the D_b of Eq. 11.

Table 4.2 shows the PTE for each multipole bin. In the case of perfect model and without correlation between the multipoles, the PTE values should appear as they were drawn from uniform distribution. Generally, the model fits the *EE* spectra well. The first two bins of the *BB* spectra seem to be a bad fit. The fact that the PTEs are low can be due to the high signal-to-noise ratio at larger scales, because in that regime the assumption of isotropy does not hold, the dust and synchrotron emissions are non-stationary and non-Gaussian, which are supposed by the SMICA likelihood. In addition the uncertainty is slightly underestimated in the likelihood for two reasons. First, the data spectra are computed on a small sky fraction, they are therefore different from the spectra computed on all possible patches with the same sky fraction. This variability is not taken into account in the likelihood. Second, the noise spectra of QUIJOTE-MFI and WMAP data are computed using half-difference maps,

Table 3. PTE per mask, mode and multipole bin for the fit of the *sfd1* model to the *WM+PI+QJ* dataset.

mask	mode	40	60	80	100	120
$\pm 10^\circ$	EE	0.450	0.278	0.302	0.454	0.001
	BB	6×10^{-5}	0.002	0.155	0.249	0.019
$\pm 15^\circ$	EE	0.114	0.192	0.648	0.335	0.006
	BB	4×10^{-5}	0.024	0.538	0.148	0.079
$\pm 20^\circ$	EE	0.145	0.137	0.737	0.186	0.024
	BB	0.002	0.024	0.752	0.172	0.071

which would correspond to one particular realisation of the noise and not the underlying true noise power spectrum. Furthermore, the first bin contains less modes and is more prone to deviate from the chi-squared distribution compared to the other bins.

5 RESULTS

This section presents the results obtained when fitting the favoured foreground model, i.e. the *sfd1* model. In the plots of this section, the points and the error bars represent the median and the 16th to 84th percentile region of the marginal posterior distribution, which correspond to a 68 % confidence level (one sigma). In this section we first compare the data and the best fit model, we then present the results on the properties of the synchrotron and more particularly the synchrotron as a foreground for CMB studies.

We tested adding the rolling of the synchrotron SED s_{run} to the modelling of the synchrotron. Since the resulting marginal posterior on s_{run} did not exclude 0, s_{run} is fixed to zero for the all the analyses presented in this paper.

5.1 Data vs. model

Fig. 3 shows qualitatively the goodness of the fit by plotting together the data cross- and auto-spectra of the *EE* data and the best fit data model for the $|b| < 10^\circ$ mask. Samples of the binned data model C_ℓ of Eq. 9 are constructed from the samples of the parameters θ . In both panels, the spectrum between two frequencies ν_1 and ν_2 is plotted against the effective frequency $\sqrt{\nu_1 \nu_2}$. The data auto-spectra are noise de-biased by subtracting the noise covariance matrix N_b of Eq. 9. For a quantitative discussion on the goodness of fit, see Sec. 4.2. The measured power of the model can be decomposed into the synchrotron, the dust, the cross power between the synchrotron and the total dust and the CMB contributions, as depicted in the right panel of Fig. 3. We note that if only the auto-spectra are taken into account, then the CMB is degenerated with the dust-synchrotron correlation at small scales, but if the cross-spectra are added to the data we do not observe any correlation between the CMB and the dust-synchrotron covariance. **The WMAP 33 data point is very low in the bin $\ell = 60$. Please check if there is a problem there.** **FV:** on the figure, the auto spectra points are the data spectra minus the estimated noise (using year 1-4 and years 5-9 as half maps to estimate noise power spectrum for WMAP). It happens that the noise estimate for WMAP 33 GHz at $\ell=60$ is rather high. **Do we have any quantitative estimate of the goodness of fit? Reduced chi2? PTE?** **FV:** I removed the plots of goodness of fit and I replaced them by the table with the PTEs (the text has been adapted)

At the largest angular scales (i.e. the lowest multipole bins, which have the highest signal-to-noise ratio), the data and the model

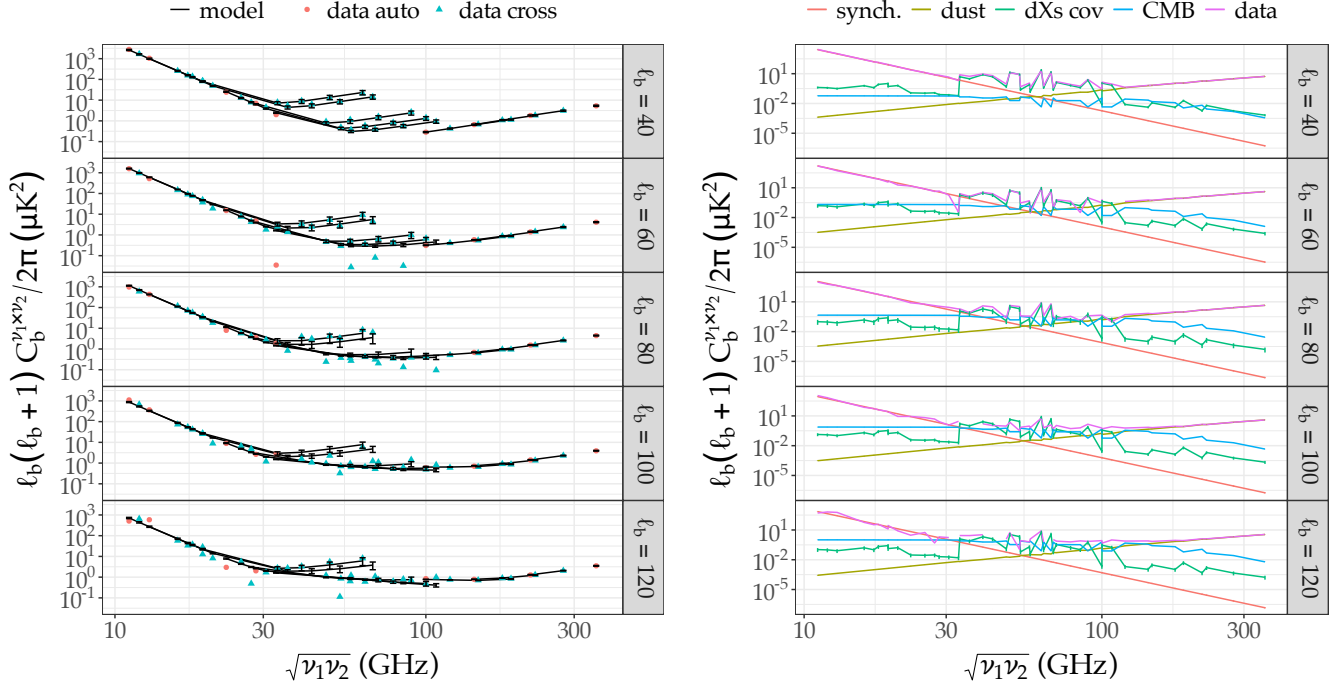


Figure 3. EE power spectra from the *WM+PI+QJ* data set and the $|b| < 10^\circ$ Galactic mask, and the best fit model in units of antenna temperature. The auto- and cross-spectra are positioned on the x-axis according to their effective frequency, e.g. a spectrum between frequency ν_1 and ν_2 is positioned at the effective frequency $\sqrt{\nu_1 \nu_2}$. The panels from top to bottom show the five multipole bins. The spectra are multiplied by a factor $\ell_b(\ell_b + 1)/2\pi$ with ℓ_b the central multipole of each bin. The noise spectra are subtracted from the data spectra. Left: noise subtracted data auto- and cross-spectra and the median best fit model with error bars representing the 15.9 to 84.1 % quantiles. The red dots and the green triangles represent the values of the auto- and cross-spectra respectively. Each black line represents the set of cross-spectra involving the same frequency, from the lowest to the highest frequency. Right: Data spectra and the break down of the model into its individual components (i.e. the sum of the components in the right panel gives the best fit model on the left plot in black) namely the synchrotron, the dust, the dust-synchrotron correlation and the CMB.

are in very good agreement. The covariance between synchrotron and dust emission, denoted as “dXs cov” in the plots, plays a major role in the fit, as nearly half of the measured cross-spectra are dominated by the dust–synchrotron correlation.

5.2 Derived synchrotron properties

Table 4 shows the best-fit values of the synchrotron spectral parameters for the *sfd1* model. The model *sfd1* is better model than the *sld1* model in the sense that the Bayes factor is in favour of the *sfd1* model (as shown in Sec. 4), but also because the distribution over all the other parameters of the model stay unchanged. We now discuss in detail these results.

5.2.1 Amplitude and angular dependence (A_s, α)

Fig. 4 shows the *EE* and *BB* synchrotron fitted spectra at 23 GHz as inferred from the three data sets (*WM+PI*, *WM+PI+QJ* and *NULL*). The numerical values of the amplitude at 23 GHz (A_s) and index α of the synchrotron power spectrum defined in equation 4 are presented in Table 4, for the three galactic masks, and the *EE* and *BB* spectra separately. The *EE* spectrum is well constrained over the entire multipole range $30 \leq \ell < 130$. The *BB* spectrum is also well determined, but most of the information of the small scales is extrapolated from the large scales. This point is also illustrated in Fig. 5, where we show that, without the parametrisation of the synchrotron spectrum of Eq. 4, the *BB* spectrum of the synchrotron above to multipole $\ell = 80$ is poorly estimated. In the smallest scales

of the *sld1* model, i.e. the two highest multipole bins, the distribution of the spectrum in the *NULL* set gives amplitudes similar to the those in the two other sets.

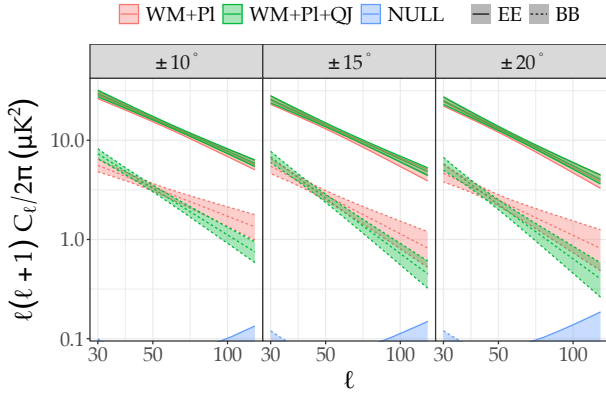
The results of the fits are in good agreement with previous works. The synchrotron *EE* power spectrum is flatter than the *BB* power spectrum, and overall the values do not agree with those reported by the S-PASS team (Krachmalnicoff et al. 2018). The discrepancy might come from the fact that the two experiment do not observe the same sky region. **JARM: to be understood.**

5.2.2 Synchrotron B-to-E ratio

Fig. 6 shows the ratio between the *BB* and *EE* synchrotron spectra, as function of multipoles. We find values between 0.1 and 0.25, with a moderate dependence on the multipole when the combination *WM+PI+QJ* is used, thanks to the reduction of the error bars on the power spectra. In the case of *WM+PI*, we do not find a clear dependence on scale, and the average value of the *B/E* ratio is approximately 0.2 for the three sky masks. These numbers are consistent with the values reported by Martire et al. (2022) using *WMAP* K-band and *Planck* LFI30 data only, which range from 0.19 ± 0.01 to 0.25 ± 0.02 for their two extreme sky masks with sky fractions 94 % and 30 %, respectively; and also with the value reported in Rubiño-Martín et al. (2023) of 0.26 ± 0.07 for $|b| > 10^\circ$ using the QUIJOTE-MFI 11 GHz map in the range $30 < \ell < 200$. Therefore, we confirm that for synchrotron emission the B-to-E power spectrum ratio is of the order of 0.2–0.25 when using QUIJOTE-MFI, *WMAP* and *Planck* data, in contrast to the values reported for ther-

Table 4. Values of the synchrotron related parameters when inferring using the *sfd1* model.

Mask	Dataset	mode	A_s [$10^{-3} \mu K^2$]	α	$\text{corr}(A_s, \alpha)$	β_s	ρ_{40}	ρ_{60}	ρ_{80}	ρ_{100}	ρ_{120}
$\pm 10^\circ$	WM+PI	EE	9.26 ± 0.59	-3.10 ± 0.12	0.69	-3.00 ± 0.09	0.21 ± 0.04	0.11 ± 0.04	0.04 ± 0.05	0.18 ± 0.05	0.14 ± 0.07
		BB	2.07 ± 0.36	-2.96 ± 0.28	0.86	-2.83 ± 0.20	0.15 ± 0.05	0.20 ± 0.06	0.08 ± 0.08	0.11 ± 0.10	0.27 ± 0.13
	WM+PI+QJ	EE	9.63 ± 0.43	-3.09 ± 0.09	0.54	-3.22 ± 0.03	0.20 ± 0.04	0.11 ± 0.04	0.08 ± 0.04	0.13 ± 0.04	0.12 ± 0.05
		BB	1.52 ± 0.22	-3.54 ± 0.22	0.86	-3.33 ± 0.07	0.16 ± 0.05	0.17 ± 0.05	0.13 ± 0.07	0.13 ± 0.09	0.16 ± 0.12
$\pm 15^\circ$	WM+PI	EE	7.62 ± 0.59	-3.17 ± 0.14	0.74	-3.05 ± 0.11	0.09 ± 0.05	0.07 ± 0.05	0.05 ± 0.05	0.18 ± 0.06	0.09 ± 0.08
		BB	1.47 ± 0.36	-3.29 ± 0.37	0.90	-3.01 ± 0.25	0.08 ± 0.06	0.19 ± 0.07	0.04 ± 0.11	0.15 ± 0.13	0.34 ± 0.19
	WM+PI+QJ	EE	8.09 ± 0.42	-3.13 ± 0.10	0.58	-3.24 ± 0.03	0.07 ± 0.05	0.07 ± 0.05	0.07 ± 0.05	0.11 ± 0.05	0.07 ± 0.06
		BB	1.06 ± 0.20	-3.84 ± 0.28	0.89	-3.34 ± 0.08	0.09 ± 0.05	0.13 ± 0.06	0.03 ± 0.09	0.14 ± 0.12	0.17 ± 0.17
$\pm 20^\circ$	WM+PI	EE	6.81 ± 0.61	-3.26 ± 0.16	0.76	-3.12 ± 0.12	0.09 ± 0.05	-0.03 ± 0.05	0.02 ± 0.06	0.17 ± 0.07	0.12 ± 0.09
		BB	1.39 ± 0.39	-3.18 ± 0.43	0.89	-3.01 ± 0.30	0.11 ± 0.07	0.19 ± 0.08	0.03 ± 0.12	0.10 ± 0.15	0.26 ± 0.21
	WM+PI+QJ	EE	7.09 ± 0.43	-3.24 ± 0.12	0.62	-3.26 ± 0.03	0.07 ± 0.05	-0.01 ± 0.05	0.04 ± 0.05	0.11 ± 0.05	0.04 ± 0.07
		BB	0.93 ± 0.22	-3.81 ± 0.34	0.90	-3.25 ± 0.10	0.09 ± 0.06	0.16 ± 0.07	-0.02 ± 0.10	0.13 ± 0.14	0.12 ± 0.20


Figure 4. Best-fit synchrotron power spectra for the *E*- and *B*-modes at 23 GHz, and for the three data sets *WM+PI* (red), *WM+PI+QJ* (green) and *NULL* (blue). The three panels show, from left to right, the three analysis masks corresponding to $|b| < 10^\circ$, 15° and 20° , respectively.

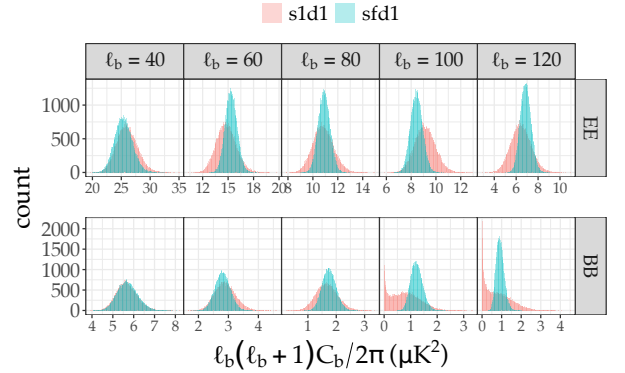
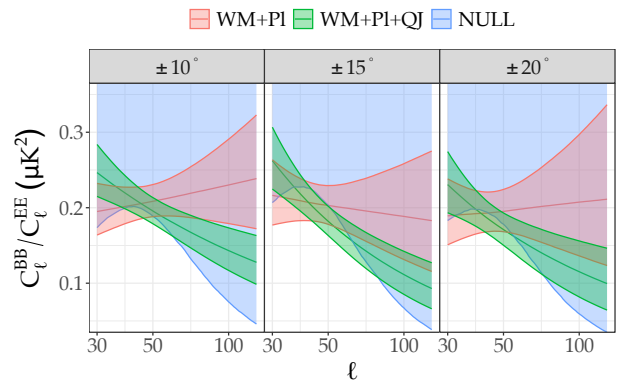
mal dust emission, which are typically of the order of 0.5 (Planck Collaboration et al. 2016, 2020b).

However, these *B/E* ratios are significantly lower than what has been reported by Krachmalnicoff et al. (2018) using S-PASS data at 2.3 GHz. Although the S-PASS and QUIJOTE-MFI sky footprints have a minimal overlap which might explain part of the difference, it is clear that the S-PASS results show a much stronger dependence with the Galactic latitude, with values ranging from 0.87 ± 0.02 for the sky mask $|b| > 20^\circ$, to 0.50 ± 0.06 for $|b| > 40^\circ$. As suggested in their paper, this is probably ascribed to some degree of Faraday rotation of the polarization angle of the signal at mid latitudes.

Finally, we observe that there is no evidence to suggest that the *EE* and *BB* spectra have different slopes. Indeed, when comparing the indices α of the *EE* and *BB* spectra (Table 4) we find a 1.9, 2.4 and 1.6 sigma difference on the $\pm 10^\circ$, $\pm 15^\circ$ and $\pm 20^\circ$ masks, respectively. **Revise FV: rephrased and removed the part about redoing the analysis and comparing evidences.**

5.2.3 Synchrotron spectral index (β_s)

Fig. 7 shows the synchrotron spectral indices β_s obtained from the fits on the three data sets. The numerical values are listed in Table 4 for the *WM+PI* and *WM+PI+QJ* datasets. The spectral index using the *NULL* dataset is centered close to -3 because of the prior distribution adopted of $\beta_s \sim \mathcal{N}(-3, 1)$. The error bars in this *NULL* case are much larger than those in the two other datasets, indicating that the data is the main source of information in the


Figure 5. Marginal distributions of the binned power spectrum of the synchrotron. The green histograms are the distribution of the binned power spectra of the *sfd1* model, i.e. with parametrization by a power, from the set *WM+PI+QJ* of Fig. 4. The red histograms are the distributions of the binned power spectra in the *s1d1* model, i.e. without parametrization by a power law. Without any model for the synchrotron spectrum, the amplitudes at each multipole bins are almost independent and there is no detection of synchrotron *BB* above $\ell \sim 80$.

Figure 6. *BB/EE* ratio of the synchrotron spectra at 23 GHz fitted on the three different data sets, and on the three galactic cuts.

estimation of β_s with both the *WM+PI* and *WM+PI+QJ* datasets. The addition of the QUIJOTE-MFI frequencies allows to reduce the error bars on β_s by a factor 2 to 3. We do not find significant differences between the *EE* and *BB* spectral index.

Our values are consistent with those works based on harmonic space analyses. Martire et al. (2022) found $\beta_s^{\text{EE}} = -2.95 \pm 0.10$ and $\beta_s^{\text{BB}} = -3.02 \pm 0.35$ for the mask with $f_{\text{sky}} = 50\%$. Similarly, we

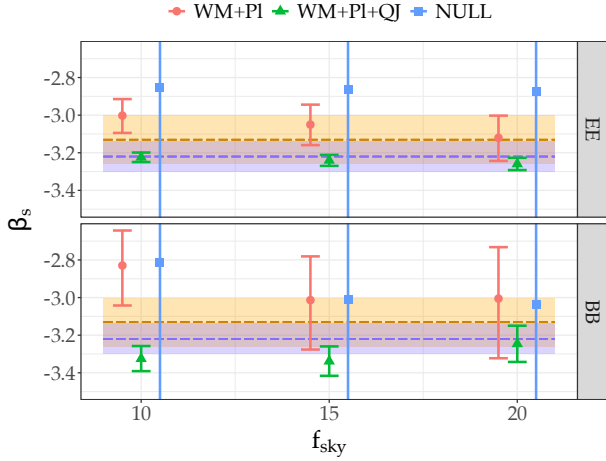


Figure 7. Synchrotron SED spectral index fitted on the three data sets, and on the three galactic cuts. The shaded areas represent the Planck and S-PASS values. Numerical values of β_s can be found in Table 4.

also find consistency with the values reported by Krachmalnicoff et al. (2018), of $\beta_s = -3.22 \pm 0.08$ for the southern sky.

The derived values are also consistent with those studies based on real-space analyses for the polarized synchrotron emission. **QUIJOTE values**

5.2.4 Correlation matrix of synchrotron parameters

Fig. 8 illustrates the correlation structure in the posterior distribution between the main synchrotron parameters, for the particular case of bin 5 (i.e. $\ell_b = 120$). We find that the amplitude of the power spectrum A_s is correlated at the level of 0.86 and 0.40 with the power spectrum index α and the synchrotron spectral index β_s , respectively. However, the correlation with the dust-synchrotron correlation parameter (ρ) is low; indeed, the value depicted in the figure of -0.1 is the largest one for all bins, with the anti-correlation decreasing down to 0 with decreasing multipoles.

5.2.5 Dust-synchrotron correlation (ρ)

Figure 9 shows the inferred dust–synchrotron correlation for the *sfd1* model, for the three data sets and the three Galactic cuts considered. We find that the correlation remains approximately constant over the multipole range and sky fractions analysed, with an average value of about 0.15, depending on the multipole bin.

The overall trend of our correlation estimates is consistent with the results reported by Krachmalnicoff et al. (2018), derived from S-PASS and *Planck* 353 GHz polarization maps (see their Fig. 12). The small differences can be explained by the variability of the dust-synchrotron correlation across the sky. Interestingly, as in our case, the correlation appears to be slightly lower in the $\ell_b = 80$ bin for the BB synchrotron spectrum.

5.3 Forecasted level of synchrotron contamination for B-mode studies

Using *sfd1* model and our best-fit results, we can build the distributions of the SED of the synchrotron and of the expected *BB* power at CMB frequencies. Fig. 10 shows the synchrotron *BB* power at 90 and 150 GHz together with the expected CMB *B*-mode amplitude in the cases where the tensor-to-scalar ratio is $r = 0.01$ or $r = 0.001$.

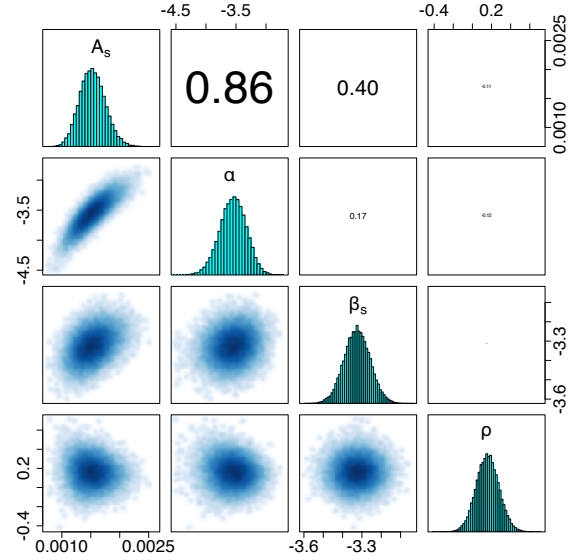


Figure 8. Example of the correlation structure of the four parameters describing the synchrotron component of the *sfd1* model: the amplitude A_s (normalized at multipole 80 and at frequency 23 GHz), the power spectrum index α , the synchrotron spectral index β_s , and the dust-synchrotron correlation ρ . We show the case of *B*-modes and the Galactic cut $|b| > 10^\circ$. The values correspond to the fifth (i.e. last) multipole bin, $\ell_b = 120$.

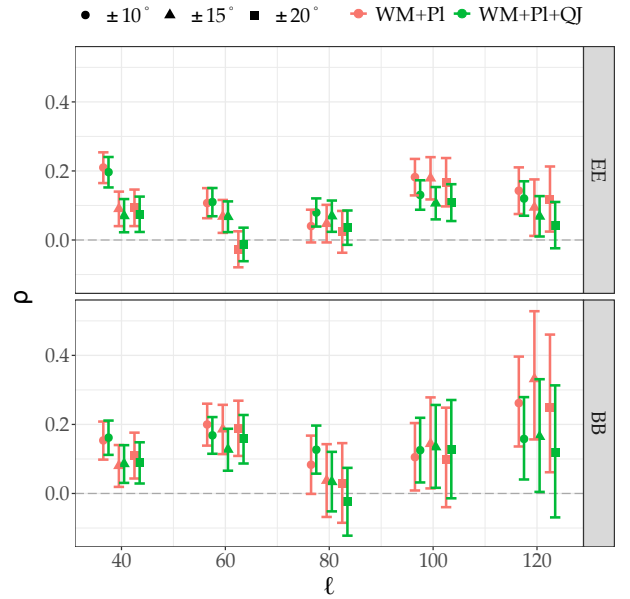


Figure 9. Dust-synchrotron correlation for E-modes (top panel) and B-modes (bottom panel), for the three Galactic cuts and for the two data sets *WM+PI* (red) and *WM+PI+QJ* (green). The *NULL* data set is not shown, as the distribution of the correlation values is nearly uniform between -1 and 1 across all multipole bins. Although all points correspond to the central multipoles $\ell_b = 40, 60, 80, 100$ and 120 , they are slightly shifted horizontally for clarity. Numerical values of ρ can be found in Table 4.

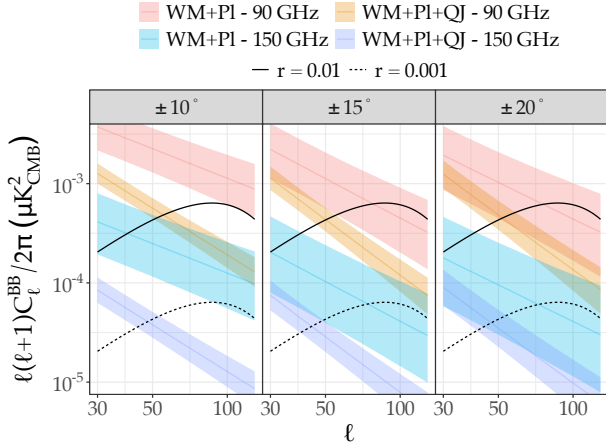


Figure 10. Contamination of the CMB at 90 and 150 GHz by the synchrotron B -modes. The shaded area represent a 1 sigma contour, i.e. between the 15.9 and 84.1 quantiles.

For completeness, Fig. 11 shows the SED of the standard deviation at the map level of the synchrotron and CMB B -modes. These standard deviations of the synchrotron and the CMB are computed using the same multipole range of the analysis ($30 \leq \ell < 120$). As for the spectra we use the fitted spectra for the synchrotron and the Planck best fit spectra so that

$$\text{SED}_{\text{synch.}} = f_\nu \left(\frac{\nu}{23\text{GHz}} \right)^{\beta_s} \sqrt{\sum_{\ell=30}^{129} \frac{2\ell+1}{4\pi} A_s \left(\frac{\ell}{80} \right)^\alpha} \quad (12)$$

$$\text{SED}_{\text{CMB}} = \sqrt{\sum_{\ell=30}^{129} \frac{2\ell+1}{4\pi} C_\ell^{\text{Planck}}}, \quad (13)$$

where f_ν is the inverse of the unit conversion factor, i.e. it converts from μK (antenna temperature) to μK_{CMB} (thermodynamic temperature). The figure shows the two cases before and after adding the QUIJOTE frequencies. **Maybe we could include the window function of a experiment with 1 deg beam, as a reference. FV: I don't see why it would bring more information, since the synchrotron and the CMB would be similarly impacted, and the aim here is to compare synchrotron with CMB.**

Given the current upper limits on r , any CMB B -mode experiment targeting intermediate angular scales and tensor-to-scalar ratios below $r = 10^{-2}$ cannot neglect the contribution of synchrotron emission in the sky masks considered in this analysis. This conclusion is consistent with the forecasts presented by Krachmalnicoff et al. (2018) for the southern sky.

6 DISCUSSION AND CONCLUSION

In the present paper we have investigated the essential properties of the synchrotron emission in polarization. We have looked into the values of some synchrotron features in light of the new QUIJOTE data. We have found no evidence for a spatial variation of the spectral index of the synchrotron β_s at the power spectrum level. However, the results of the model selection might not be valid in the general case. Even though QUIJOTE alone do not detect the variability of synchrotron index, a comparison of synchrotron spectral index values with those of S-PASS and Planck shows the variability of β_s across the sky. We find a value of the BB/EE ratio different from that of previous analyses. Again, this is to put on the fact that the

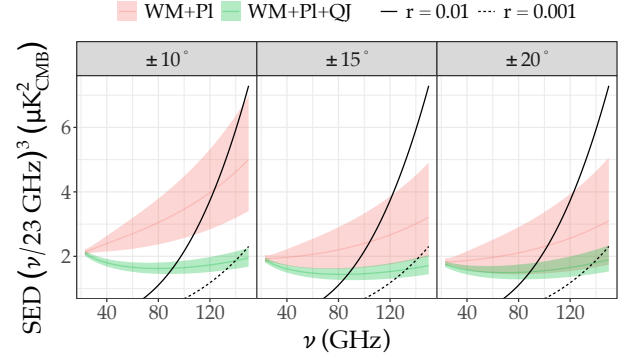


Figure 11. Spectral energy distribution of the standard deviation of the synchrotron B -modes, for two sets $WM+Pl$ and $WM+Pl+QJ$, and the three galactic cuts. The contours represent the 1 sigma contour, i.e. between the 15.9 and 84.1 quantiles of the posterior distribution. The black lines represent the CMB SED for tensor-to-scalar ratio 0.01 and 0.001. We note that the SEDs are multiplied by a factor ν^3 for readability purposes.

properties of the diffuse synchrotron is not constant over the sky. Our results tend to show that the EE spectrum is flatter than the BB spectrum on the QUIJOTE region. The S-PASS region exhibits the opposite property and with more significance. Concerning the spatial correlation with the dust, we found that the cross-correlation for both the E - and B -modes is mostly between 0.1 and 0.25. We have also investigated the level of contamination to CMB B -modes due to synchrotron. On the region an multipole range of the analysis, we find that the synchrotron emission is on average equivalent to $r = 0.01$ at 90 GHz and to $r = 0.001$ at 150 GHz. The current and future CMB B -mode experiment target at least a 2σ detection of $r = 0.01$. Thus the analysis for the detection of the CMB B -modes must include at least one low frequency channel to mitigate the contamination due to the synchrotron.

In the present study, we neglect Faraday depolarization in the modeling of the data. This effect slightly decreases the synchrotron power by an amount that depends on the frequency. The SED of the auto- and cross-spectra should then be different from the usual power law with spectral index β_s . This effect could explain part of the slightly bad fit of the E -modes in Fig. 4.2.

We supposed that only synchrotron and dust emissions contribute to the polarization of the data. The fact that the single component model is preferred by the data to model the synchrotron emission indicates that neither the free-free emission nor the AME is polarized to a detectable level. A study must be conducted in order to put a quantitative upper bound on the polarization fraction of the free-free emission and the AME. This can be achieved by adding to the modeling of the data an extra component with an unconstrained SED. A quantitative upper bound could then be derived from the posterior distribution on the power of the extra component at each frequency.

In this study we mask a $b = \pm 10^\circ$ band around the galactic plane. The analyses we conducted on this region should be done again but after masking the galactic plane with a largest galactic mask. This would allow to investigate the evolution of the properties of the synchrotron with the galactic latitude.

We leave the considerations of the previous paragraphs to future studies.

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Parameter	Mean	Standard deviation	Median	1 σ levels	2 σ levels	Parameter	Mean	Standard deviation	Median	1 σ levels	2 σ levels
Asf[1,1]	1.946	0.239	1.503	1.708	1.934	model_cov[1,1,1]	4.564	0.239	4.121	4.326	4.552
Asf[1,2]	0.762	0.083	0.608	0.679	0.758	model_cov[1,1,2]	2.264	0.157	1.968	2.107	2.258
Asf[1,3]	0.006	0.001	0.005	0.005	0.006	model_cov[1,1,3]	0.105	0.009	0.088	0.096	0.105
Asf[1,4]	0.022	0.002	0.018	0.020	0.022	model_cov[1,1,4]	0.207	0.019	0.171	0.188	0.206
Asf[1,5]	0.002	0.000	0.002	0.002	0.002	model_cov[1,1,5]	0.066	0.006	0.056	0.061	0.066
Asf[2,1]	0.454	0.064	0.335	0.390	0.451	model_cov[1,1,6]	0.008	0.002	0.004	0.006	0.008
Asf[2,2]	0.178	0.024	0.134	0.154	0.177	model_cov[1,1,7]	0.011	0.003	0.004	0.007	0.011
Asf[2,3]	0.001	0.000	0.001	0.001	0.001	model_cov[1,1,8]	0.017	0.006	0.007	0.012	0.017
Asf[2,4]	0.005	0.001	0.004	0.004	0.005	model_cov[1,1,9]	0.030	0.010	0.011	0.021	0.030
Asf[3,1]	0.152	0.039	0.080	0.114	0.151	model_cov[1,2,1]	2.264	0.157	1.968	2.107	2.258
Asf[3,2]	0.060	0.015	0.032	0.045	0.059	model_cov[1,2,2]	2.798	0.083	2.644	2.715	2.794
Asf[3,4]	0.002	0.000	0.001	0.001	0.002	model_cov[1,2,3]	0.066	0.006	0.055	0.060	0.065
Asf[4,1]	0.101	0.034	0.035	0.069	0.101	model_cov[1,2,4]	0.130	0.012	0.108	0.118	0.129
Asf[4,2]	0.040	0.013	0.014	0.027	0.040	model_cov[1,2,5]	0.042	0.004	0.035	0.038	0.041
Asf[4,4]	0.001	0.000	0.000	0.001	0.001	model_cov[1,2,6]	0.005	0.001	0.003	0.004	0.005
Asf[5,1]	0.085	0.034	0.014	0.051	0.085	model_cov[1,2,7]	0.007	0.002	0.003	0.005	0.007
Asf[5,2]	0.033	0.013	0.006	0.020	0.033	model_cov[1,2,8]	0.011	0.003	0.004	0.007	0.011
Adf[1,7]	0.002	0.000	0.002	0.002	0.002	model_cov[1,2,9]	0.019	0.006	0.007	0.013	0.019
Adf[1,8]	0.005	0.000	0.005	0.005	0.005	model_cov[1,3,1]	0.105	0.009	0.088	0.096	0.105
Adf[1,9]	0.017	0.001	0.015	0.016	0.017	model_cov[1,3,2]	0.066	0.006	0.055	0.060	0.065
Adf[2,8]	0.002	0.000	0.002	0.002	0.002	model_cov[1,3,3]	0.014	0.001	0.013	0.014	0.014
Adf[2,9]	0.007	0.000	0.006	0.006	0.007	model_cov[1,3,4]	0.011	0.001	0.009	0.010	0.011
Adf[3,8]	0.001	0.000	0.001	0.001	0.001	model_cov[1,3,5]	0.004	0.000	0.003	0.003	0.004
Adf[3,9]	0.004	0.000	0.003	0.003	0.004	model_cov[1,3,8]	0.001	0.000	0.001	0.001	0.001
Adf[4,9]	0.002	0.000	0.002	0.002	0.002	model_cov[1,3,9]	0.002	0.001	0.001	0.002	0.002
Adf[5,9]	0.001	0.000	0.001	0.001	0.001	model_cov[1,4,1]	0.207	0.019	0.171	0.188	0.206
betas	-3.114	0.072	-3.259	-3.187	-3.115	model_cov[1,4,2]	0.130	0.012	0.108	0.118	0.129
betad	1.552	0.006	1.540	1.546	1.552	model_cov[1,4,3]	0.011	0.001	0.009	0.010	0.011
rs1d1[1]	0.166	0.051	0.063	0.115	0.166	model_cov[1,4,4]	0.041	0.002	0.038	0.040	0.041
rs1d1[2]	0.177	0.055	0.065	0.121	0.177	model_cov[1,4,5]	0.007	0.001	0.006	0.006	0.007
rs1d1[3]	0.133	0.075	-0.016	0.059	0.131	model_cov[1,4,7]	0.001	0.000	0.001	0.001	0.001
rs1d1[4]	0.110	0.090	-0.053	0.025	0.106	model_cov[1,4,8]	0.002	0.001	0.001	0.001	0.002
rs1d1[5]	0.117	0.110	-0.059	0.026	0.110	model_cov[1,4,9]	0.004	0.001	0.002	0.003	0.004
r13[1]	0.453	0.034	0.383	0.419	0.454	model_cov[1,5,1]	0.066	0.006	0.056	0.061	0.066
r13[2]	0.412	0.030	0.350	0.382	0.413	model_cov[1,5,2]	0.042	0.004	0.035	0.038	0.041
r13[3]	0.361	0.027	0.305	0.334	0.362	model_cov[1,5,3]	0.004	0.000	0.003	0.003	0.004
r13[4]	0.312	0.025	0.261	0.287	0.312	model_cov[1,5,4]	0.007	0.001	0.006	0.006	0.007
r13[5]	0.347	0.022	0.302	0.325	0.347	model_cov[1,5,5]	0.022	0.000	0.022	0.022	0.022
						model_cov[1,5,9]	0.002	0.000	0.001	0.001	0.002
						model_cov[1,6,1]	0.008	0.002	0.004	0.006	0.008
						model_cov[1,6,2]	0.005	0.001	0.003	0.004	0.005
						model_cov[1,6,6]	0.001	0.000	0.001	0.001	0.001
						model_cov[1,6,7]	0.001	0.000	0.001	0.001	0.001
						model_cov[1,6,8]	0.002	0.000	0.002	0.002	0.002
						model_cov[1,6,9]	0.003	0.000	0.003	0.003	0.003
						model_cov[1,7,1]	0.011	0.003	0.004	0.007	0.011
						model_cov[1,7,2]	0.007	0.002	0.003	0.005	0.007
						model_cov[1,7,4]	0.001	0.000	0.001	0.001	0.001
						model_cov[1,7,6]	0.001	0.000	0.001	0.001	0.001
						model_cov[1,7,7]	0.002	0.000	0.002	0.002	0.002
						model_cov[1,7,8]	0.003	0.000	0.003	0.003	0.003
						model_cov[1,7,9]	0.006	0.000	0.005	0.005	0.006
						model_cov[1,8,1]	0.017	0.006	0.007	0.012	0.017
						model_cov[1,8,2]	0.011	0.003	0.004	0.007	0.011
						model_cov[1,8,3]	0.001	0.000	0.001	0.001	0.001
						model_cov[1,8,4]	0.002	0.001	0.001	0.001	0.002
						model_cov[1,8,6]	0.002	0.000	0.002	0.002	0.002
						model_cov[1,8,7]	0.003	0.000	0.003	0.003	0.003
						model_cov[1,8,8]	0.006	0.000	0.005	0.005	0.006
						model_cov[1,8,9]	0.010	0.001	0.009	0.009	0.010
						model_cov[1,9,1]	0.030	0.010	0.011	0.021	0.030
						model_cov[1,9,2]	0.019	0.006	0.007	0.013	0.019
						model_cov[1,9,3]	0.002	0.001	0.001	0.002	0.002
						model_cov[1,9,4]	0.004	0.001	0.002	0.003	0.004
						model_cov[1,9,5]	0.002	0.000	0.001	0.001	0.002
						model_cov[1,9,6]	0.003	0.000	0.003	0.003	0.003
						model_cov[1,9,7]	0.006	0.000	0.005	0.005	0.006
						model_cov[1,9,8]	0.010	0.001	0.009	0.009	0.010
						model_cov[1,9,9]	0.017	0.001	0.015	0.016	0.017
						model_cov[2,1,1]	2.373	0.064	2.254	2.309	2.369
						model_cov[2,1,2]	0.987	0.063	0.859	0.923	0.987
						model_cov[2,1,3]	0.025	0.003	0.019	0.022	0.024

Table 5. Caption